

COPD

Model card with explicit reference parameters

This card records the model using an explicit reference parameter set. The purpose is to stop the model card from depending on hidden defaults inside the model JSON.

Identity

Model	COPD
Reference parameter set	reference-parameter-set:ncd-copd:synthetic-v1
Scenario registry	scenario-registry:ncd-copd:reference-synthetic-v1
Source model hash	d600016bdb6240e5

Provenance

The current parameter snapshot copies 189 generated values from the source model. It includes 22 external data-backed values and 158 static single-value assumptions. 19 country-bound parameters are made explicit in the synthetic reference set.

Internal validity

The current-reference test materialises the model from the snapshot and runs it for Ethiopia. The synthetic-reference test materialises the model with the synthetic reference country where possible. These tests are execution checks, not external scientific validation.

Current-reference run	pass
Current-reference rows	12320
Synthetic-reference run	pass
Synthetic-reference rows	12320

Face validity

The headline table below uses the synthetic reference fixture run. This is a plausibility prompt for the explicit-reference machinery, not a claim that the synthetic reference parameters are externally validated.

Headline outputs

Measure	2025	2030	2040
Population at risk	979,167	974,868	961,394
COPD prevalence state	20,767	24,577	31,932
COPD incidence flow	784	780	770
COPD mortality flow	16	19	25
Healthy years lived	997,492	996,576	990,370

Limits

This is the first conservative migration. It preserves non-country embedded parameters and makes the reference-set machinery explicit. The next stronger version should replace disease, risk-factor, and cost inputs with declared synthetic fixture values where those are intended to be stable across evidence updates.